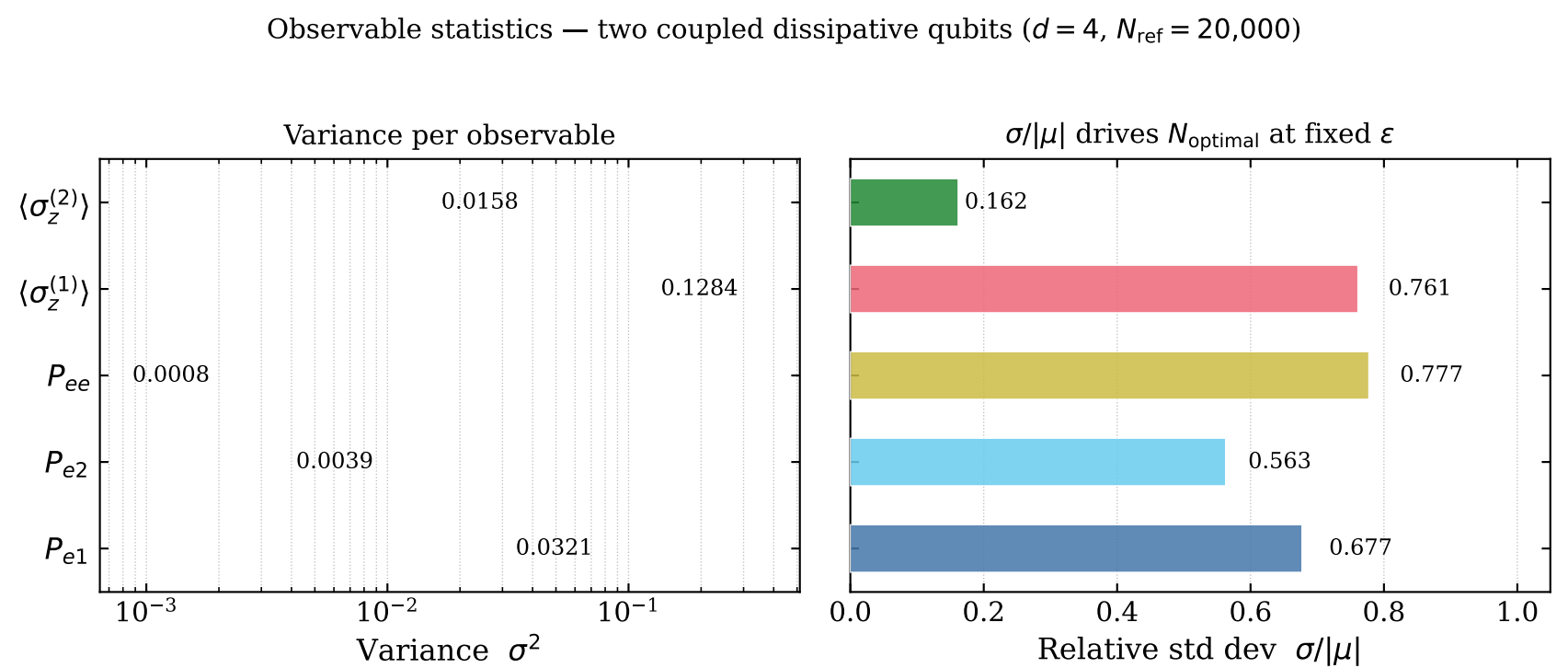
Observable statistics — two coupled dissipative qubits ($d = 4$, $N_{\mathrm{ref}} = 20{,}000$)
Variance per observable
$\langle\sigma_z^{(2)}\rangle$
$\langle\sigma_z^{(1)}\rangle$
P_{ee}
P_{e2}
P_{e1}
0.0158
0.1284
0.0008
0.0039
0.0321
10^{-3}
10^{-2}
10^{-1}
Variance σ^2
$\sigma/|\mu|$ drives N_{optimal} at fixed ε
0.162
0.761
0.777
0.563
0.677
0.0
0.2
0.4
0.6
0.8
1.0
Relative std dev $\sigma/|\mu|$
$\langle\sigma_z^{(2)}\rangle$ sits near its steady-state floor ⇒ small $\sigma/|\mu|$ ⇒ far fewer trajectories needed.